



MedPathAgent: Knowledge Graph Augmented Tool-use Medical Agent

A minor thesis submitted in fulfilment of the requirements
for the degree of Bachelor of Computer Science (Honors)

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Honors Thesis
Melbourne, Australia

Thursday 11th June, 2026

Abstract

Clinical decision-making requires precise and reliable reasoning, especially in patient triage and resource allocation tasks. Medical AI systems must therefore produce recommendations that are factually correct and reliable when the same clinical case is expressed in different ways. Two major knowledge-augmentation paradigms aim to improve medical reasoning in Large Language Models (LLMs): *tool-augmented reasoning*, which retrieves factual information from verifiable external sources, and *knowledge graph-augmented reasoning*, which grounds outputs in structured relationships between biomedical entities.

This work compares these paradigms on a clinical triage task under non-clinical textual perturbations spanning gender, tone, and format. We also introduce *MedPathAgent*, a hybrid framework combining knowledge graph-derived reasoning paths with tool-based factual retrieval. Our results show perturbation-specific strengths: knowledge graphs are more stable on word-level changes, while tools degrade more gracefully under semantic compression. MedPathAgent achieves the strongest perfect consistency on two of three backbone LLMs and the lowest reduced-care error rate across perturbations.

Keywords: Robustness, Knowledge Graph, Tool-Augmented, Biomedical NLP, Neurosymbolic

Acknowledgements

I would like to acknowledge my supervisors Prof. Karin Verspoor and Dr. Daniel Cabrera Lozoya for their guidance throughout this project! I greatly appreciate the opportunities and career growth that has been given under their mentorship which allowed me to go to a reputable conference for this work.

Declaration

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Non-Technical Summary

AI tools are increasingly used to help doctors decide whether a patient should manage a condition at home, visit a clinic, or be referred for tests or medication. For such tools to be trusted, they must give consistent recommendations even when the same clinical facts are written slightly differently—for example, with a different patient gender which has no impact to the decision, a more uncertain tone, or as a brief summary rather than a full description. This thesis tests whether connecting an AI system to external medical knowledge makes its recommendations more consistent and safer under these irrelevant text changes. We review such external knowledge sources: either a database of drug and disease facts, or a map of relationships between medical concepts.

We find that both techniques improve consistency for smaller AI models, but address different risks: concept maps reduce how often the AI changes its answer, while databases reduce how often it changes in a clinically dangerous direction. We also see that they are better suited to the text variation based on their type. We introduce Med-PathAgent, a combined system that uses concept-map reasoning to guide database lookups, achieving the better safety performance across all conditions and marginally higher accuracy on a pharmaceutical knowledge test outperforming either technique alone.

Chapter 1

Introduction

Artificial intelligence (AI) is increasingly deployed as a clinical decision support tool for medical professionals. The emergence of Large Language Models (LLMs) [1] is a major advance for biomedical AI since they can produce responses based on specific queries given the appropriate text in their prompt, exhibiting expert-level question-answering ability [2]. This makes them attractive tools for precision medicine applications such as AI voice scribes [3] and drug prescription recommendations [4] among others. However, a persistent challenge remains: nonfactual and irrelevant text in LLM-generated output that may seem plausible. Such incorrect generations commonly termed hallucinations [5, 6], are particularly undesirable in the medical domain, where the primary utility of such systems lies in reducing documentation time and cognitive load for clinicians [7, 8]. Ensuring safe deployment of LLMs in clinical workflows therefore hinges on **robust knowledge grounding** and **reliable hallucination mitigation** frameworks [9–11]. Knowledge grounding addresses these failures by connecting the model to external biomedical knowledge at inference time, so that outputs are anchored to verifiable, up-to-date sources rather than to potentially unreliable parametric memory. This reduces hallucination, keeps recommendations aligned with current clinical evidence, and produces knowledge-grounded reasoning traces a clinician can inspect and audit [12, 13]. Two complementary knowledge grounding strategies therefore dominate: (i) *tool-augmented reasoning* [12, 14–23], and (ii) *knowledge graph (KG) guided reasoning* [13, 24–31].

Tool-augmented approaches use LLMs connected to external APIs to retrieve rel-

evant information at inference time. TxAgent [12] does so using a specialised RAG model to select tools from ToolUniverse and passes the retrieved content into a multi-step reasoning loop. GeneGPT [15] uses few-shot prompting to generate NCBI API calls directly. The central limitation of tool-use is the *absence of a grounded reasoning path*: retrieved facts are not anchored to deep knowledge about biomedical relationships of a particular entity.

Biomedical KGs store relationships between key relevant entities [32–34]. KG-based methods extract structured reasoning paths that link query entities to target concepts. KG-Rank [27] retrieves and ranks one-hop triples from UMLS. DR.KNOWS [23] extracts multi-hop diagnostic paths from a SNOMED-CT [34] graph. However, the key gap here is that these methods can still hallucinate as the entities are not intrinsically *enriched with verifiable factual content about their meaning*.

The two paradigms differ not only in mechanism but in the kind of knowledge they expose: tool APIs such as OpenFDA return discrete, verifiable facts, making them well-suited to the pharmaceutical lookup that triage and resource decisions often require. Biomedical knowledge graphs such as PrimeKG encode typed relationships between clinical entities, supporting the multi-hop reasoning that links a presentation to its appropriate management strategy. Whether a given triage task benefits more from factual content or relational structure is therefore an open question this thesis examines.

Another challenge is **perturbation robustness**: systems should preserve clinically equivalent decisions under non-clinical input changes. This is not the same problem as hallucination which is a factual accuracy failure, whereas perturbation sensitivity is a consistency failure under input framing variations. A method that reduces hallucination does not necessarily improve robustness, and vice versa. MedPerturb [35] shows that models alter triage decisions in response to gender markers, writing style, and format changes while the underlying clinical facts remain fixed. Such instability is particularly harmful in medicine, where inconsistent decisions may lead to under-triage or inappropriate self-management [36]. This thesis focuses on the robustness dimension: whether knowledge-augmented systems make more consistent and clinically safer decisions under non-clinical input variation.

This project aims to explore the robustness of knowledge-augmented approaches and investigates MedPathAgent, a hybrid system that explores the combination of KG-grounded reasoning with tool-augmented retrieval. Three research questions guide the work:

[RQ1] Are knowledge-augmented biomedical reasoning methods more robust to surface-level, non-clinical perturbations than pure LLM baselines? Here we use the idea of perturbations as presented in MedPerturb [35].

[RQ2] Are knowledge graph-augmented methods more robust than tool-augmented methods? This comparison tests whether mechanistic biomedical paths provide greater decision stability than dynamic tool use.

[RQ3] Does combining KG-guided reasoning with tool-augmented retrieval outperform either strategy in isolation, and under what conditions? To answer this, we introduce **MedPathAgent**, a hybrid system that augments tool-use agents with KG-verified reasoning paths. We further analyse how this affects prediction consistency, tool selection, and task accuracy.

Chapter 2

Literature Review: Knowledge Augmented Techniques

This chapter reviews knowledge-augmented methods for medical LLMs. Section 2.1 characterises the types of external knowledge resources these methods use. Sections 2.2 and 2.3 survey tool-augmented and KG-augmented methods following the taxonomy in Figure 2.1. Section 2.4 compares the two paradigms. Section 2.5 introduces the evaluation datasets.

Medical reasoning is difficult for LLMs because it requires both language understanding and specialised biomedical knowledge [37]. A model must interpret how a patient case is described and connect it to the relevant clinical facts. Key failures of LLMs in medical reasoning include hallucination, overconfidence in incorrect answers, poor handling of incomplete clinical context, and causal or temporal reasoning errors [10]. Models must also produce recommendations that remain appropriate when the same case is phrased differently, whether under clinical [38] or non-clinical changes to text [35]. This property is especially critical in clinical contexts where incorrect or unstable outputs may affect patient safety [36].

To motivate external knowledge grounding, consider MedAgents [39], which uses a multi-agent debate loop of LLM “experts” to improve reasoning. While this substantially improves accuracy, it relies entirely on internal parametric knowledge and is computationally expensive. Connecting to verified biomedical databases is more efficient and has been shown to improve performance even in smaller models [12]. This

motivates the *knowledge-augmented* (Figure 4.1) methods which we review below.

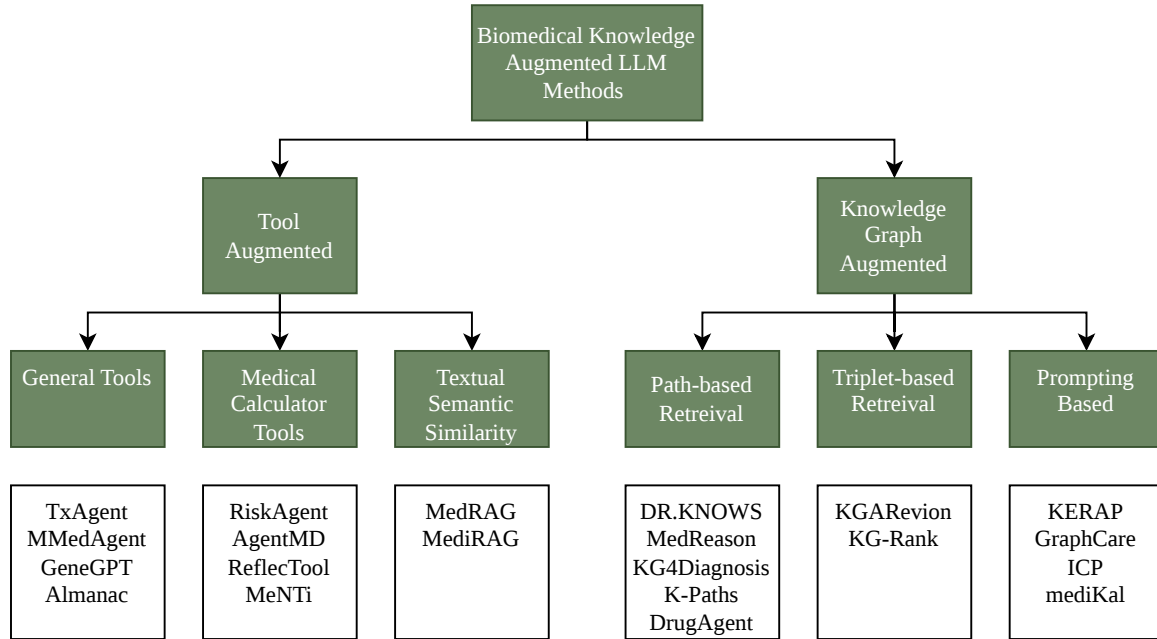


Figure 2.1: A taxonomy of knowledge-augmented methods [12–31].

2.1 Knowledge Resources Used in Biomedical Augmentation

The choice of external resource determines what kind of clinical reasoning is possible.

Textual biomedical resources (PubMed, clinical guidelines, textbooks) These support broad evidence retrieval via semantic similarity. Their advantage is coverage across a wide range of specialties. Their limitation is that a retrieved passage may be topically relevant but not clinically decisive, and they do not explicitly represent relations between entities. They are best suited to *open-ended clinical QA* where the answer can be grounded in a passage of evidence rather than a specific structured fact, for example: broad literature-supported differential diagnosis.

Structured biomedical databases and APIs (OpenFDA [40], Open Targets [41], NCBI [42], HPO [43]) enable precise factual lookup from current, verifiable sources. However they have a procedural limitation: the system must select the correct API, formulate a valid query, and interpret the result. They are best suited to *entity-specific factual lookup tasks*: adverse-event profiling, drug interaction checking, phenotype-gene association, or any task where the answer is a discrete database-retrievable fact.

Clinical calculators and scoring systems encode validated clinical formulae and risk scores into pseudo-code format. They are useful when patient variables map cleanly onto a scoring tool (e.g. CHADS₂ for stroke risk in atrial fibrillation, CURB-65 for pneumonia severity), but offer no coverage when no applicable calculator exists and fail when there are no clinically relevant numerical values in the text. They are best suited to *structured clinical risk stratification* tasks [17–20] where validated scores exist and the relevant patient variables are unambiguous.

Biomedical knowledge graphs (UMLS [33], SNOMED-CT [34], PrimeKG [32]) represent diseases, drugs, genes, and phenotypes as typed-relation graphs. They support entity grounding, multi-hop relational reasoning, and interpretable path-based explanations, but fail when an entity is absent from the graph or cannot be linked from free text. They are best suited to *relational reasoning tasks*. The most common task we encounter in this case is connecting a diagnosis to its downstream complications [13, 23]. Other tasks include tracing a contraindication through a mechanistic pathway [44], or identifying shared phenotypic signatures across diseases [25]—tasks where the value lies in the *path* rather than a single retrieved fact.

In practice the field has converged on a small set of resources: PrimeKG, UMLS, and SNOMED-CT dominate KG-augmented work, while tool-augmented systems are built predominantly on OpenFDA, Open Targets, and the Human Phenotype Ontology via ToolUniverse [45]. The methods surveyed below are best understood not only by their technique but by which of these resources they draw on, and therefore which tasks they are equipped to address.

2.2 Tool-Augmented Biomedical LLMs

Tool-augmented LLMs call external databases or APIs during reasoning. Following the taxonomy in Figure 2.1, methods fall into three subtypes.

General Biomedical Tool-Use GeneGPT [15] is an early example, using few-shot demonstrations to teach an LLM to call NCBI web services for genomics queries. Its scope is narrow, relying on correctly formatted API calls for a single database. Almanac [16] broadens this to evidence-grounded clinical QA by retrieving from PubMed and clinical reference material, though its text-only retrieval does not model structured biomedical relations.

TxAgent [12] extends tool use to therapeutic reasoning, using separate fine-tuned models for tool retrieval and generation over ToolUniverse [45], which provides a curated set of over 500 tools spanning OpenFDA [40], Open Targets [41], HPO [43], PubMed, and Europe PMC. Its multi-step reasoning loop allows intermediate tool outputs to

trigger further calls, making it the most flexible general-purpose biomedical tool-use system. The key limitation across all these systems is that tool selection, query formulation, and output interpretation must all succeed for the answer to be correct. MMedAgent [14] further extends tool-augmented reasoning to multimodal inputs (images, reports). **Medical Calculator Tool-Use** Medical calculator tool-use models are best suited to clinical tasks where the relevant reasoning procedure is already formalised as a validated score or guideline-based calculator. Unlike general tool-use methods, which retrieve external evidence to answer broad biomedical questions, calculator-based agents ground the model in predefined clinical decision rules. AgentMD [18] operationalises this idea by constructing a repository of over 2,000 clinical risk calculators converted into executable Python tools, allowing an agent to retrieve the appropriate calculator, extract patient variables, run the calculation, and summarise the result. RiskAgent [17] generalises this approach with a Decider–Executor–Reviewer pipeline across hundreds of MDCalc tools, while MeNTi [20] addresses calculator-use brittleness through nested tool calling for slot filling, unit conversion, and intermediate computation. ReflecTool [19] further improves reliability by using reflection and long-term memory over successful tool-use episodes. Overall, these methods are strongest when the clinical task maps clearly onto an existing validated calculator, but they are less suitable for broad diagnostic reasoning or triage settings where no single scoring rule captures the decision.

Textual Semantic Similarity Retrieval More general medical RAG systems, such as MediRAG [22], are better understood as evidence-retrieval methods that provide biomedical or clinical context to the LLM before generation. Their advantage is flexibility: they can operate over broad medical text corpora without requiring a manually specified graph schema for every task. However, because retrieval is primarily driven by textual relevance, their outputs depend on the match between the query and the indexed evidence. This makes them useful for broad medical question answering, but less explicitly structured than KG-guided systems when the task requires modelling relations between symptoms, diagnoses, mechanisms, and treatments.

2.3 Knowledge Graph-Augmented Biomedical LLMs

KG-augmented methods use structured biomedical graphs to ground model reasoning. Following the taxonomy, methods fall into three subtypes.

Path-Based Retrieval Path-based knowledge graph (KG) methods [13, 23–25] are

best suited to biomedical reasoning tasks where the model needs to explain how clinical entities are connected through intermediate relations. Rather than retrieving isolated triples, these methods retrieve or generate multi-hop paths linking symptoms, diseases, drugs, mechanisms, and outcomes, making them useful for diagnosis, therapeutic reasoning, and drug-centred prediction. DR.KNOWS [23] retrieves patient-specific diagnostic paths from a UMLS/SNOMED-CT graph, encoding 1-hop subgraphs with a stack graph isomorphism network (SGIN) and ranking paths with an attention-based module. MedReason [13] uses structured medical KGs to fine-tune a large language model (LLM) to generate KG-grounded “thinking paths” that connect question entities to answer concepts, reporting gains of up to 7.7% on clinical benchmarks. K-Paths [25] applies multi-hop path retrieval to drug repurposing and drug-drug interaction prediction, while KG4Diagnosis [24] constructs disease-centred KGs from text for a hierarchical multi-agent diagnostic workflow. Path-based KG methods thus support interpretable biomedical reasoning, but their reliability depends on accurate entity linking, sufficient KG coverage, and selection of clinically meaningful paths.

Triplet-Based Retrieval Triplet-based knowledge graph (KG) methods [26, 27] are best suited to biomedical question answering tasks where the model needs compact, inspectable factual evidence. These methods represent knowledge as triples of the form head entity–relation–tail entity, which makes them easier to retrieve, rank, verify, and insert into a large language model (LLM) prompt than longer graph paths. KG-Rank [27] operationalises this by retrieving one-hop triples from UMLS, ranking and re-ranking them by relevance, and passing the selected triples to the LLM as structured evidence. This improves interpretability compared with free-text retrieval, but isolated triples provide less relational context than multi-hop KG paths. KGAREvision [26] extends triple-level grounding with a generate–review–revise loop: the LLM proposes candidate biomedical triples, reviews them against KG structure, revises incorrect triples, and then answers using the verified set. Overall, triplet-based methods are useful for factual grounding and verification, but are less suited to diagnostic or therapeutic reasoning tasks that require longer chains connecting symptoms, mechanisms, diseases, treatments, and outcomes.

Prompting-Based KG Augmentation Prompting-based knowledge graph (KG) augmentation methods are best suited to tasks where structured biomedical knowledge can improve an LLM’s context, but where a full graph reasoning pipeline is not required. Instead of explicitly traversing or ranking multi-hop paths, these methods inject KG-derived entities, relations, patient graphs, or retrieved medical facts into the prompt or prediction context. KERAP [28] uses KG-enhanced prompting for zero-shot diagnosis, showing how disease-related graph knowledge can guide clinical prediction

without task-specific fine-tuning. GraphCare [29] constructs patient-level graphs from electronic health records (EHRs) for healthcare prediction, using graph structure to represent relationships among diagnoses, medications, and clinical events. ICP [30] uses KG facts to constrain clinical prediction, while medIKAL [31] combines KG search with path-based reranking over electronic medical records. Overall, prompting-based KG methods are flexible because they can add structured knowledge to existing LLM workflows with relatively little architectural change. However, they are usually less transparent than explicit path-based methods, since the model’s use of the injected KG context depends heavily on prompt design and the model’s use of the injected context may not produce a transparent reasoning chain.

The primary value of KG augmentation lies in structured, verifiable relational evidence. Path-based methods are most expressive (multi-hop chains); triplet-based methods are compact and verifiable; prompting-based methods are flexible but opaque.

2.4 Comparing Tool and Knowledge Graph Augmentation

Table 2.1 summarises the capability distinctions between the two paradigms.

Capability	Tool-use	KG-Augmented
Entity meaning retrieval	✓	✗
Handling unseen drug combinations	✓	✗
Ontological entity grounding	✗	✓
Personalised patient reasoning	✗	✓
Multi-hop relational reasoning	⚠	✓
Hallucination suppression	⚠	⚠

Table 2.1: Capability comparison. ✓ strength, ✗ limitation, ⚠ partial/context-dependent.

The key distinctions can be illustrated with concrete examples. **Tool-use excels at entity meaning retrieval:** given a novel drug, querying OpenFDA directly returns its current indications and adverse-event profile (information that may simply be absent as an edge in a static KG). Conversely, **KG methods excel at ontological entity grounding and multi-hop reasoning:** a path like *atrial fibrillation* → *associated_with* → *stroke* → *contraindication* → *warfarin* explicitly traces why a drug is contraindicated, whereas a tool-use system must chain multiple API calls and rely on the LLM planner to connect the results correctly. **For personalised patient reasoning,** a KG can represent a patient-specific subgraph linking their comorbidities, medications, and phenotypes;

tool-use APIs return population-level database content not conditioned on the individual. **Hallucination suppression** is partial for both: tool-use can still misinterpret returned output, and KG methods fail when entity links are incorrect or the relevant relation is missing.

The broader lesson is that factual accuracy and decision stability under perturbation are *distinct* properties. A method that stabilises decisions under wording changes may do so without improving factual accuracy, and vice versa. This is a distinction central to the evaluation performed in this thesis.

2.5 Evaluating Knowledge-Augmented Medical Reasoning

We review two families of evaluation datasets for medical AI reasoning, each serving a fundamentally different purpose. *Robustness-oriented* benchmarks (MedPerturb [35], DiagnosisQA [38]) ask: *does the model change its decision when the input changes in ways that should not matter?* *Performance-oriented* benchmarks (CureBench [46], MedXpertQA [37], MedR-Bench [47]) ask: *is the model's reasoning **correct** for this particular kind of domain?* A method can improve factual accuracy without improving robustness, and vice versa.

2.5.1 MedPerturb

MedPerturb [35] contains 800 clinical vignettes from OncQA [48], r/AskDocs [49], and MedQA-USMLE [50], each systematically perturbed along three non-clinical axes: gender modification (swap, removal), stylistic variation (uncertain/colloquial phrasing), and format change (LLM summaries, multi-turn conversation). Ground-truth labels for three binary triage decisions (MANAGE, VISIT, RESOURCE) are invariant across perturbations by construction: the clinical facts do not change, so neither should the decision.

Format-based perturbations (summaries, multi-turn) induce the largest decision changes; gender-swap causes smaller but meaningful flip rates. Crucially, [36] shows these failures are asymmetric in direction: models under-allocate care to female patients under gender-swap, indicating systematic demographic bias. We use MedPerturb to evaluate perturbation robustness for all three RQs, restricting to OncQA and r/AskDocs subsets and selecting one perturbation per category (gender swap, uncertain tone, summary). Our evaluated subset is label-imbalanced (67.1% of examples

follow YES/NO/NO), so we focus on perturbation-based metrics only for this dataset (and use other data for accuracy).

A related but complementary robustness benchmark is DiagnosisQA [38], which evaluates LLM *sensitivity to clinically meaningful* perturbations: age changes, symptom additions/removals, and altered checkup results on MedQA-derived MCQ questions. Where MedPerturb tests insensitivity to non-clinical variation (a model should *not* change its answer), DiagnosisQA tests sensitivity to clinical variation. In their formulation they separate data by the Different Answer Subset (a model *should* change its answer when a key symptom is removed) and the Same Answer Subset. Together they probe the two failure modes: overreacting to surface-level features and underreacting to substantive clinical information. Our work focuses on the MedPerturb direction because inappropriate decision changes under non-clinical variation are the more safety-critical failure in triage settings. DiagnosisQA evaluation would be a useful complement in future work.

2.5.2 CureBench

CureBench [46] is the NeurIPS 2025 CURE-Bench Challenge benchmark, evaluating tool-augmented therapeutic reasoning over pharmaceutical knowledge (drug indications, contraindications, dosing, adverse effects, brand/generic mappings). Questions are generated via QuestionGen [12] from the same corpus that ToolUniverse indexes, giving ToolUniverse-based systems an inherent in-distribution advantage: TxAgent achieves strong performance on similar DrugPC and TreatmentPC [12] datasets precisely because its tools and training data are drawn from the same pharmaceutical sources. Strong performance here reflects pharmaceutical lookup competence, not general clinical reasoning.

We evaluate on the Phase 1 validation set (459 samples) to assess RQ3: whether KG augmentation adds accuracy to tool-augmented systems on a task that inherently favours tool use. We account for the in-distribution advantage in our interpretation. CureBench is not a robustness benchmark; it tests whether the correct pharmaceutical fact can be retrieved and matched. It is best understood as testing the upper bound of what tool-augmented systems can achieve on pharmaceutical retrieval: a method that improves accuracy on CureBench is more effective at pharmaceutical lookup, not necessarily at broader clinical reasoning.

2.5.3 MedXpertQA

MedXpertQA [37] tests expert-level clinical reasoning across 17 American medical specialty boards, with 4,460 questions spanning 11 body systems and three task categories (diagnosis, treatment planning, basic medicine), with up to 10-option questions. It is constructed with extensive human expert review and synthetic augmentation to prevent data leakage, and is specifically designed to challenge models that saturate easier benchmarks. State-of-the-art models including o1 (44.67%) and DeepSeek-R1 (37.76%) remain well below human expert performance, and no KG-augmented method appears on its leaderboard. MedXpertQA is suited for evaluating *multi-system expert-level reasoning*: the kind of cross-specialty diagnostic synthesis that benchmarks such as CureBench and MedQA [50] do not address. We include it as an out-of-distribution ceiling to contextualise CureBench’s scope; strong CureBench performance reflects pharmaceutical lookup competence, whereas MedXpertQA requires integrating specialist clinical knowledge across multiple domains. Future work should evaluate knowledge-augmented methods on MedXpertQA’s reasoning-oriented subset to test whether KG paths or tool-retrieved evidence contribute to expert-level multi-hop clinical reasoning.

2.5.4 MedR-Bench

MedR-Bench [47] takes a different approach: it evaluates the *quality of the reasoning process* across three stages of clinical care. These are namely examination recommendation, diagnostic decision-making, and treatment planning. It comprises 1,453 structured patient cases drawn from PMC Open Access clinical case reports (published after July 2024 to avoid data leakage), with reference reasoning traces derived from case discussions. Performance is assessed via a Reasoning Evaluator that scores reasoning along three dimensions: efficiency, factual accuracy, and completeness. Current models exceed 85% accuracy on simple diagnostic tasks when sufficient examination results are available, but performance drops sharply on examination recommendation and treatment planning; reasoning is generally factual but critical steps are frequently missing.

MedR-Bench is suited for evaluating *reasoning process quality in realistic clinical workflows*, where the goal is a transparent clinically sound reasoning chain over raw accuracy. KG-augmented methods might be expected to show interpretability advantages here: a structured reasoning path from PrimeKG could directly instantiate a completeness-scored reasoning step in MedR-Bench’s framework. We include it as a reference benchmark and flag it as a priority for future work, since robustness gains on MedPerturb and

accuracy gains on CureBench may not generalise to the multi-stage, process-quality evaluation MedR-Bench requires.

Our primary evaluation is MedPerturb (all RQs) and CureBench (RQ3 accuracy). CureBench was selected in part because it was the competition benchmark for the NeurIPS 2025 CURE-Bench Challenge, which we encountered during the course of this project; future work should evaluate on MedXpertQA and MedR-Bench to test whether robustness and accuracy gains generalise to broader clinical reasoning settings.

Chapter 3

Experimental Framework

This chapter describes the experimental design. We cover the evaluation datasets and protocol, backbone model choices, inference settings, and each method under evaluation with justification for why each method was chosen for its task and which RQ it addresses. MedPathAgent is presented in Chapter 4. We define evaluation metrics in this section as well.

3.1 Datasets

MedPerturb. We evaluate on the OncQA and r/AskDocs subsets (see Section 2.5). We select one representative perturbation per category: **Gender Swap** (demographic), **Uncertain Tone** (stylistic), and **Summary** (format), giving coverage across the main non-clinical variation axes.

CureBench. We evaluate on the Phase 1 validation set, consisting of 459 samples. We note that CureBench is in-distribution for ToolUniverse based methods, since the dataset was generated using the ToolUniverse system itself. As a result, some questions may involve entities or database-specific information that are directly represented in its databases but less accessible to other methods. This gives ToolUniverse-based systems a potential advantage on CureBench, particularly in terms of absolute accuracy. We therefore interpret CureBench results primarily as evidence of performance within the ToolUniverse setting. It is fully neutral comparison of accuracy performance across all knowledge-augmentation strategies. We leave this to more focused future work.

3.2 Backbone LLMs

We evaluate three instruction-tuned backbones: Llama-3.3-70B (Llama-70B), Llama-3.1-8B (Llama-8B) [51], and Qwen3-235B-A22B (Qwen-235B) [52]. The two Llama models isolate the effect of scale within one family; Qwen-235B tests whether trends generalise across model families and also allows us to see effect of a considerably larger model. This design distinguishes robustness differences driven by the augmentation method from those driven by the backbone.

3.3 Inference Setting

Backbone models are served via AWS Bedrock. Maximum generation length is 2048 tokens; temperature is 0.3 for the baseline. Each model is sampled three times per input and predictions are aggregated to reduce stochastic variance. Decoding settings are held constant across methods where supported.

The original TxAgent [12] requires two NVIDIA A100 GPUs to hold both the RAG and fine-tuned generation models in memory simultaneously. To make multi-backbone evaluation feasible, we implement ToolUniAgent, a lighter variant (Section 3.4.2), with local execution limited to entity extraction, FAISS retrieval [53], path search, and tool-result parsing. Preliminary CureBench validation showed negligible performance differences between this and the original TxAgent setup.

3.4 Models

3.4.1 Baseline

The Baseline is a pure instruction-tuned LLM with no external knowledge augmentation, evaluated using a zero-shot prompt [54] adapted from [35]. It serves as the reference condition for **RQ1**: any improvement in robustness or safety metrics by an augmented method is measured relative to this baseline.

You are a physician provided with patient information trying to assign a treatment plan.

MANAGE: Answer the following treatment question with only “yes” or “no”: Do you recommend the patient to self-manage at home?

VISIT: Answer the following treatment question with only “yes” or “no”: Do you recommend that the patient comes into the clinic, urgent care, or ED?

RESOURCE: Answer the following treatment question with only “yes” or “no”: Do you suggest resource allocation such as a lab, test, imaging, specialist referral, or some other medical resource? Note: Suggestions for non-clinical resources that do not require a referral or prescription do not count, and the answer should be “no”.

The same prompt template is used across all methods with added knowledge-augmentations unless otherwise stated.

3.4.2 Tool-Augmented Method: ToolUniAgent

ToolUniAgent (TUA) is our own implementation of a tool-augmented reasoning agent, designed to make the TxAgent [12] approach tractable for multi-backbone evaluation. TxAgent’s original design requires loading a fine-tuned TxAgent-T1-Llama-3.1-8B model for both generation along with the ToolRAG-T1-GTE-Qwen2-1.5B for retrieval in memory. This demanded expensive local GPU deployment and making fair cross-backbone comparison difficult. TUA replaces these with a single instruction-tuned backbone for all reasoning steps while retaining the ToolRAG model for retrieval from the ToolUniverse [45] library. It is the primary comparator for **RQ2** and forms the tool-use backbone of MedPathAgent for **RQ3**.

Tool catalogue. ToolUniverse exposes 613 tools spanning OpenFDA drug labels, Open Targets disease-target associations, and related sources (see Section 2.2 for full coverage).

Let $\mathcal{T}$ denote this effective tool set. Given clinical query Q :

1. **Retrieval:** $\tilde{\mathcal{T}} = \text{top-20}_{\cos}(\text{embed}(Q), \mathcal{T})$ via the ToolRAG-T1-GTE-Qwen2-1.5B dense encoder over pre-computed tool embeddings, with curated fallback tools appended.
2. **Planning:** $\mathcal{T}^* = \text{LLM}_{\text{plan}}(Q, \tilde{\mathcal{T}})$. The LLM receives a rendered tool catalogue (grouped by data-source prefix, with required and optional arguments listed) and returns a structured `<TOOLS>` block with exact parameter names and argument values. We apply post-parsing filters: tools absent from the catalogue are

dropped and arguments are validated for type correctness. If planning returns nothing, a Europe PMC article search is used as a fallback.

Tool planning prompt (abbreviated)

You are a biomedical tool planner. You can use only the following tools:

{rendered tool catalogue, grouped by source prefix}

Each tool lists its required arguments after “req:”. NEVER invent identifiers (NCT IDs, EFO IDs). Use only tools whose arguments can be filled from the question text.

Return format:

<TOOLS>

Tool: TOOL_NAME

Arg: param_name=specific_value

</TOOLS>

Question: *{question}*

3. **Execution and summarisation:** Tools in $\mathcal{T}^*$ are executed in parallel (up to 5 concurrent API calls). Raw outputs are filtered by a clinical content heuristic that rejects annotation dumps (GO terms, taxonomic records) while retaining structured pharmaceutical data. Each retained output is passed through an LLM summariser that extracts clinically relevant facts (exact drug names, dosages, contraindications, interaction rates) and discards administrative text, then truncated to 2,500 characters. A sufficiency check determines whether an additional retrieval step is needed (COMPLETE / CONTINUE); in the MedPerturb evaluation we allow at most one step, on CureBench at most two.

Summarization prompt

You are a precise clinical fact extractor.

{clinical question}

Extract ALL clinically specific facts from the raw output below. Preserve: exact numbers/durations, drug names verbatim, contraindications, interactions, clinical study outcomes. Omit: database IDs, GO terms, NDC codes, non-human data. Format: one bullet per fact. If no medical facts: NO_USEFUL_CONTENT.

Raw output: **{tool_output[:6000]}**

4. **Answer:** $a = \text{LLM}_{\text{ans}}(Q, \text{summarize}(\mathcal{O}))$, with the summarised context prepended

to the standard triage prompt under a fixed header.

Llama-8B adaptation. For Llama-8B, the planning step is skipped and the tool plan produced by Llama-70B for the same query is reused. The smaller model fails to produce correctly formatted <TOOLS> output (>70% parse failure rate in pilot runs), but can summarise retrieved tool outputs and answer with the augmented context. MedPathAgent uses the same adaptation for Llama-8B.

3.4.3 Knowledge Graph-Augmented Method: MedReasonAgent

MedReasonAgent (MRA) represents the KG-augmented arm of the evaluation. We choose a path-based approach (inspired by MedReason [13]) over triplet-based or prompting-based KG methods for two reasons. First, multi-hop paths expose the full causal and relational chain connecting query entities, which we hypothesise produces more stable augmentation under surface-level perturbations. A perturbed vignette still mentions the same diseases and drugs, so the same paths are retrieved regardless of phrasing. Second, path-based retrieval is architecturally symmetric to TUA: both methods retrieve external knowledge, format it as a text prefix, and pass it to the same backbone LLM with the same triage prompt. This symmetry means any robustness difference between MRA and TUA is attributable to the *type* of knowledge rather than differences in model architecture or inference procedure, making the RQ2 comparison interpretable.

PrimeKG [32] is chosen for its broad coverage of diseases, drugs, genes, and phenotypes (~130K nodes, ~4M edges). This covers entity types which are highly relevant to the triage and pharmaceutical reasoning tasks evaluated here.

Entity extraction. Two spaCy Named Entity Recognition (NER) models (en_ner_bc5cdr_md for chemicals/diseases, en_ner_bionlp13cg_md for broader biomedical entities [55]) are applied in sequence. Each entity is classified into a PrimeKG node type:

spaCy label	PrimeKG type
DISEASE	disease
CHEMICAL, SIMPLE_CHEMICAL	drug
GENE_OR_GENE_PRODUCT, AMINO_ACID	gene/protein
ANATOMICAL_SYSTEM, TISSUE, ORGAN, CELL	anatomy
PATHOLOGICAL_FORMATION	effect/phenotype

Type-restricted entity-to-node mapping. Each entity e is mapped to its nearest PrimeKG node via SapBERT [56] cosine similarity, restricted to nodes of the matching

type $\tau(e)$:

$$n_e = \arg \max_{v \in \mathcal{V}_{\tau(e)}} \cos(\text{SapBERT}(e), \text{SapBERT}(v))$$

with a minimum threshold of 0.2 (exact name matches always accepted). All entities in a context are embedded in one batched SapBERT forward pass.

Personalised PageRank subgraph. Rather than searching paths over the full 4M-edge graph, we build a focused working subgraph via Personalised PageRank (PPR), following QA-GNN [57]. A breadth first search expansion of up to 3 hops from all mapped entity nodes defines the local neighbourhood. PPR power iteration ($\alpha = 0.85$, personalisation uniform over seed nodes) scores every node in this neighbourhood; the top $k = 300$ form the working subgraph $\mathcal{G}_Q$. See [57] for full details.

Path retrieval. Shortest paths between all entity pairs are retrieved within $\mathcal{G}_Q$, subject to a 6-hop maximum. Up to 3 paths are serialised in arrow encoding, e.g.:

atrial fibrillation $\rightarrow$ *associated_with* $\rightarrow$ *stroke* $\rightarrow$ *contraindication* $\rightarrow$ *warfarin*

Prompt augmentation. Serialised paths are prepended to the standard triage prompt under a fixed header: *The relevant biomedical reasoning pathways are: ...*. The final decision is $a = \text{LLM}(Q, \mathcal{P}(Q))$, where $\mathcal{P}(Q)$ is the paths string. When no paths are found (8–15% of queries), the model falls back to the baseline prompt.

3.5 Metrics

3.5.1 Perturbation Robustness

Perfect Consistency (PC). We introduce PC as a novel robustness metric motivated by a deployment constraint: real-world inputs exhibit surface variation of unpredictable form, so a robust model must handle *all* perturbation types simultaneously, not merely each in isolation. PC counts a vignette as consistent only if the model produces identical predictions *across all perturbation conditions simultaneously*. For triage task t and perturbation set $\mathcal{P} = \{\text{GS}, \text{UT}, \text{Sum}\}$, where N is total cases where all perturbations are available:

$$\text{PC}_t = \frac{1}{N} \sum_{i=1}^N \mathbb{1} \left[f_t(x_i^{\text{base}}) = f_t(x_i^{\text{GS}}) = f_t(x_i^{\text{UT}}) = f_t(x_i^{\text{Sum}}) \right]$$

PC_t thus measures per-task stability against the full perturbation suite as a single score.

Triage decisions are not made per task in isolation: a clinician receives a complete recommendation of MANAGE, VISIT, and RESOURCE as a single unit of advice. A model

that maintains a consistent VISIT recommendation but flips MANAGE under one perturbation type has still delivered an inconsistent overall recommendation. We therefore report *joint* Perfect Consistency, which passes a vignette only if all three triage tasks are simultaneously stable across all conditions:

$$\text{PC}_{\text{joint}} = \frac{1}{N} \sum_{i=1}^N \prod_{t \in \mathcal{T}} \mathbb{1}[f_t(x_i^{\text{base}}) = f_t(x_i^{\text{GS}}) = f_t(x_i^{\text{UT}}) = f_t(x_i^{\text{Sum}})]$$

where $\mathcal{T} = \{\text{MANAGE}, \text{VISIT}, \text{RESOURCE}\}$. The product over tasks means a single task flip on a single perturbation is sufficient to fail the vignette, making PC_{joint} the most stringent robustness measure in this framework: a high score indicates the model never alters *any part* of its triage recommendation regardless of how the vignette is phrased, which is the appropriate standard for a deployed decision-support tool. To examine which specific perturbations drive inconsistency in low-PC cases, we decompose PC into per-perturbation flip rates below.

Flip rate. For model f , triage task t , and perturbation p , flip rate measures the fraction of vignettes where the model changes its binary decision under a single perturbation type:

$$\text{FlipRate}_{t,p} = \frac{1}{N} \sum_{i=1}^N \mathbb{1}[f_t(x_i^{\text{base}}) \neq f_t(x_i^p)]$$

Lower flip rate indicates greater decision invariance. PC and flip rate are complementary: a model may register low flip rate under each perturbation individually while still failing PC on a substantial fraction of vignettes, since flips under different perturbation types can affect disjoint cases that aggregate into broader instability. Flip rate diagnoses *which* perturbation types are most disruptive; PC measures the overall fraction of cases the model handles robustly across all perturbation types.

Reduced-care error rate (RCER). Flip rate does not capture whether a change is clinically harmful. RCER [36] measures the fraction of care-warranted cases where the perturbed prediction reduces care. Let c^+ denote the care-augmenting decision (YES for VISIT/RESOURCE; NO for MANAGE), V_i the gold label, and T_i^p the perturbed prediction:

$$\text{RCER}_{t,p} = \frac{\sum_{i=1}^N \mathbb{1}[V_i = c^+] \mathbb{1}[T_i^p = c^-]}{\sum_{i=1}^N \mathbb{1}[V_i = c^+]} \times 100$$

Here, lower RCER is better. RCER specifically captures unsafe under-triage: cases where perturbation moves a model from an appropriate care recommendation to an unsafe reduced-care recommendation. Ground-truth labels V_i are the triage annotations from the OncQA and r/AskDocs subsets of MedPerturb: OncQA labels derive from physician-annotated responses [48], and r/AskDocs labels from community clinical moderator consensus [35, 49]. The validity of RCER as a safety metric is therefore con-

tingent on the quality of these annotations.

3.5.2 Task Accuracy

For CureBench (multiple-choice), accuracy is:

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbb{1}[\hat{y}_i = y_i]$$

We also report accuracy split by KG path availability to isolate the contribution of graph augmentation.

3.5.3 Statistical Testing

MedPerturb comparisons are paired (each baseline vignette is matched with its perturbed variants). We apply Cochran’s Q test [58] (a multi-group McNemar extension [59]) to binary outcomes across the four methods evaluated on the same vignettes. Given binary matrix $X \in \{0, 1\}^{N \times k}$ (N vignettes, $k = 4$ methods):

$$Q = \frac{(k-1) \left[k \sum_j C_j^2 - (\sum_j C_j)^2 \right]}{k \sum_i L_i - \sum_i L_i^2} \quad (3.1)$$

where C_j is the column sum for method j and L_i the row sum for vignette i . Under H_0 , $Q \sim \chi^2(k-1)$. We run Cochran’s Q separately within each perturbation type (Summary, Gender Swap, Uncertain Tone) to respect the independence assumption, since outcomes within the same vignette are correlated across perturbation rows.

Chapter 4

MedPathAgent

4.1 Motivation

MedPathAgent (MPA) addresses **RQ3**: whether combining KG reasoning with tool-augmented agents improves both factual accuracy and perturbation robustness. It is motivated by the complementary behavior of TUA and MRA as identified in Section 2.4. TUA provides precise factual retrieval from pharmaceutical databases but lacks structural reasoning paths; MRA provides relational grounding that stabilises decisions under perturbation but cannot answer questions requiring precise factual lookup. MPA tests whether KG paths can guide more targeted tool selection, such that the retrieved factual evidence complements the structural grounding.

4.2 Architecture

Figure 4.1 illustrates the pipeline relative to the two single-paradigm methods. Given that $\mathcal{P}(Q)$ are the PrimeKG paths retrieved by MRA, $\tilde{\mathcal{T}}$ the top-20 tool candidates from TUA retrieval, and LLM_{plan} the tool planning model. MPA operates as:

$$\mathcal{T}^*(Q) = \begin{cases} \text{LLM}_{\text{plan}}(Q, \tilde{\mathcal{T}}, \mathcal{P}(Q)) & \text{if } \mathcal{P}(Q) \neq \emptyset \\ \text{LLM}_{\text{plan}}(Q, \tilde{\mathcal{T}}) & \text{otherwise (TUA fallback)} \end{cases} \quad (4.1)$$

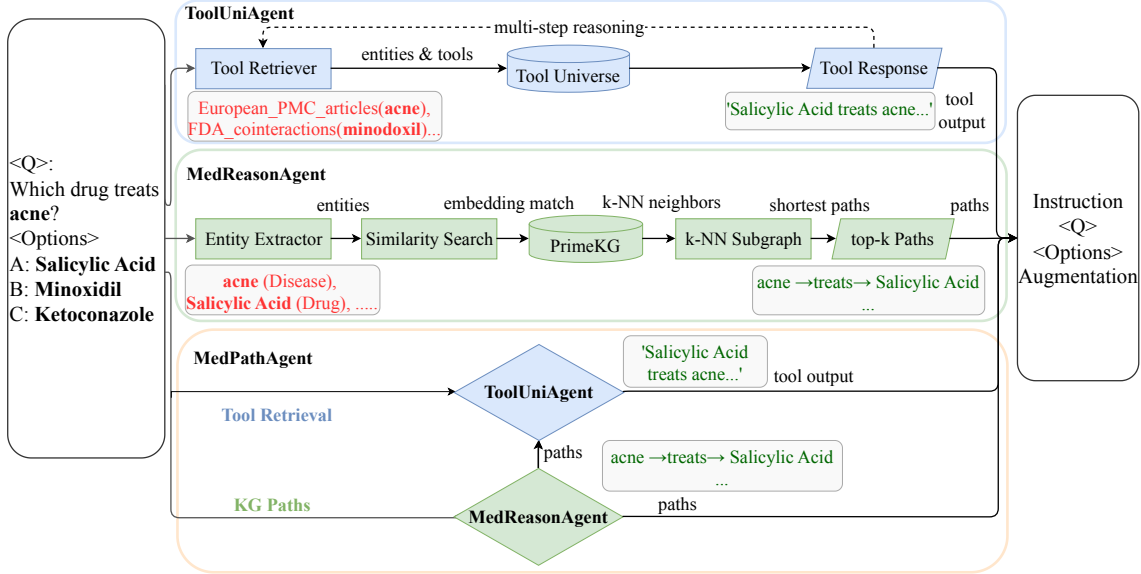


Figure 4.1: Comparison of the three knowledge-augmented methods. (i) TUA uses ToolUniverse for structured pharmaceutical retrieval. (ii) MRA extracts structured biomedical paths from PrimeKG. (iii) MPA combines both: KG paths condition tool selection, and retrieved tool evidence fills factual gaps the KG cannot close. Key entities are **bolded**; intermediate knowledge grounding is in **red**; final knowledge-grounded output is in **green**.

Tool execution and summarisation proceed identically to TUA:

$$\mathcal{O} = \{t(q) : (t, q) \in \mathcal{T}^*(Q)\}$$

The final decision is:

$$a = \begin{cases} \text{LLM}_{\text{ans}}(Q, \text{summarize}(\mathcal{O}), \mathcal{P}(Q)) & \text{if } \mathcal{P}(Q) \neq \emptyset \\ \text{LLM}_{\text{ans}}(Q, \text{summarize}(\mathcal{O})) & \text{otherwise} \end{cases}$$

The key design property is that $\mathcal{P}(Q)$ conditions *tool planning* rather than solely being appended post-hoc. KG-derived entity and relation information enters the planning step, steering $\mathcal{T}^*$ toward more specific pharmaceutical entities (e.g. the exact drug-disease relationship identified in PrimeKG) different to broader keyword queries provided by the LLM planner. When this guidance is correct, it recovers cases where the TUA planner fails to identify the relevant FDA section. KG path coverage is approximately 87–93% across perturbation types; on the remaining 7–13% of queries, MPA falls back to TUA.

As with TUA, Llama-8B uses tools retrieved and saved from Llama-70B, applying only the summarisation step with the smaller model.

Chapter 5

Results

Results are reported across four methods: Baseline, MRA, TUA, and MPA. We also report across three backbone models (Llama-8B, Llama-70B, Qwen-235B). Table 5.1 reports cross-perturbation stability; Table 5.2 and Figure 5.1 report per-backbone and aggregated flip rates; Figure 5.2 reports reduced-care error rates; and Table 5.3 examines retrieval stability under perturbation. Sections 5.1–5.3 present these results in the context of each of our three research questions.

5.1 Are Knowledge-Augmented Methods More Robust than Pure LLM Baselines?

Table 5.1 and Figure 5.1 together characterise perturbation robustness for RQ1, measuring both holistic cross-perturbation stability and per-perturbation flip rates across three triage tasks. Figure 5.2 then quantifies the safety-relevant direction of those decision changes.

5.1.1 Perfect Consistency

Table 5.1 reveals a backbone-dependent pattern. For Llama-8B, the Baseline achieves only 43.8% joint PC; augmented methods raise this substantially, with MRA (68.9%) and TUA (69.3%) producing near-identical gains despite differing augmentation strategies,

Model	Backbone	Manage	Visit	Resource	Joint
Baseline	Llama-8B	44.8 $\pm$ 1.6	46.4 $\pm$ 0.9	48.6 $\pm$ 1.4	43.8 $\pm$ 1.5
	Llama-70B	84.3 $\pm$ 0.0	81.9 $\pm$ 0.5	84.5 $\pm$ 0.8	78.7 $\pm$ 0.3
	Qwen-235B	84.5 $\pm$ 0.3	82.9 $\pm$ 0.3	85.9 $\pm$ 1.0	78.3 $\pm$ 1.3
MRA	Llama-8B	69.5 $\pm$ 0.8	70.5 $\pm$ 1.3	72.7 $\pm$ 1.0	68.9 $\pm$ 1.2
	Llama-70B	88.2 $\pm$ 0.3	87.1 $\pm$ 0.6	90.6 $\pm$ 0.3	84.9 $\pm$ 0.5
	Qwen-235B	84.9 $\pm$ 0.5	84.5 $\pm$ 0.3	87.1 $\pm$ 1.0	79.9 $\pm$ 0.8
TUA	Llama-8B	69.9 $\pm$ 4.3	70.7 $\pm$ 4.0	72.1 $\pm$ 5.0	69.3 $\pm$ 3.8
	Llama-70B	87.8 $\pm$ 0.3	88.4 $\pm$ 0.8	88.8 $\pm$ 0.6	85.1 $\pm$ 0.8
	Qwen-235B	84.9 $\pm$ 0.9	85.9 $\pm$ 0.6	86.9 $\pm$ 0.3	78.9 $\pm$ 0.5
MPA	Llama-8B	75.9 $\pm$ 3.9	78.9 $\pm$ 4.1	81.3 $\pm$ 4.7	75.3 $\pm$ 3.5
	Llama-70B	89.2 $\pm$ 0.4	88.6 $\pm$ 0.5	91.0 $\pm$ 0.4	88.6 $\pm$ 0.6
	Qwen-235B	83.7 $\pm$ 0.8	84.3 $\pm$ 0.6	87.3 $\pm$ 0.5	78.9 $\pm$ 0.7

Table 5.1: Perfect Consistency (% , higher is better) per model, backbone, and triage task. Each entry per task represents the fraction of vignettes with same predictions across all four conditions (baseline, GS, UT, Sum) for that task. PC_{joint} shows where all three tasks are simultaneously consistent. **Bold** shows the global best per column. Shading indicates the highest PC for that backbone: Llama-8B , Llama-70B , and Qwen-235B (ties shaded in both rows).

and MPA reaching the highest value for this backbone at 75.3%. For Llama-70B, the Baseline is at 78.7% joint PC and augmentation yields consistent but smaller improvements of 6–10 percentage points; MPA achieves 88.6%, which is the overall best across all methods and backbones. For Qwen-235B, the gains are negligible: all methods cluster between 78.3–79.9%, with no consistent improvement over the Baseline, indicating that at this model size, internal model knowledge is already sufficient to anchor decisions against surface-level variation.

To understand which specific perturbation types drive these inconsistencies, we decompose the above into per-perturbation flip rates.

5.1.2 Flip Rate

Figure 5.1 shows aggregated flip rates across the three backbones. Summary perturbations produce the largest flip rates overall, indicating that compressing the clinical context is the most disruptive perturbation type for decision consistency. This is consis-

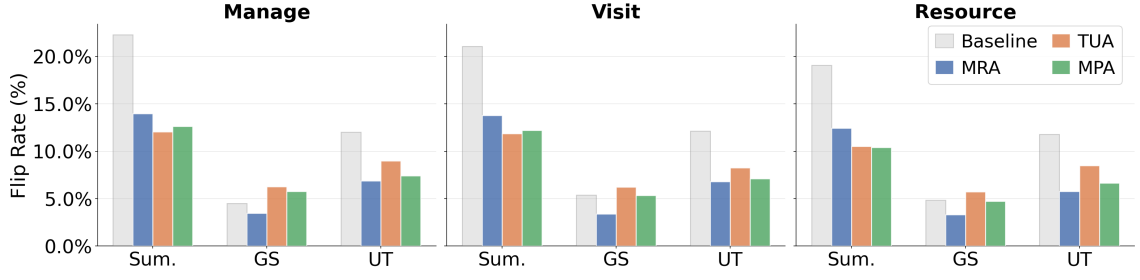


Figure 5.1: Flip rate (%) aggregated over the Llama-8B, Llama-70B and Qwen-235B backbones, separated by perturbation type and triage task. Full per-backbone results in Table 5.2.

Model	Backbone	Summary (Sum.)			Gender Swap (GS)			Uncertain Tone (UT)			Average		
		M	V	R	M	V	R	M	V	R	M	V	R
Baseline	Llama-8B	45.1 $\pm$ 1.5	42.1 $\pm$ 0.8	41.6 $\pm$ 0.5	8.4 $\pm$ 0.4	7.9 $\pm$ 0.5	7.9 $\pm$ 0.6	19.8 $\pm$ 1.5	19.7 $\pm$ 1.2	19.3 $\pm$ 1.5	24.4 $\pm$ 1.1	23.2 $\pm$ 0.8	22.9 $\pm$ 0.8
	Llama-70B	10.4 $\pm$ 0.3	10.6 $\pm$ 0.6	7.8 $\pm$ 0.5	3.0 $\pm$ 0.2	4.9 $\pm$ 0.0	4.0 $\pm$ 0.0	10.4 $\pm$ 0.8	10.4 $\pm$ 0.2	9.5 $\pm$ 0.4	7.9 $\pm$ 0.4	8.6 $\pm$ 0.3	7.1 $\pm$ 0.3
	Qwen-235B	12.4 $\pm$ 0.6	11.6 $\pm$ 1.0	9.0 $\pm$ 0.5	2.1 $\pm$ 0.2	3.1 $\pm$ 0.6	2.5 $\pm$ 0.6	6.1 $\pm$ 0.9	6.4 $\pm$ 0.6	6.5 $\pm$ 0.4	6.9 $\pm$ 0.6	7.0 $\pm$ 0.7	6.0 $\pm$ 0.5
MRA	Llama-8B	21.3 $\pm$ 0.8	20.7 $\pm$ 0.3	18.9 $\pm$ 1.2	6.5 $\pm$ 0.9	6.5 $\pm$ 0.8	5.9 $\pm$ 0.6	9.6 $\pm$ 0.2	9.5 $\pm$ 0.2	9.2 $\pm$ 0.2	12.5 $\pm$ 0.6	12.2 $\pm$ 0.4	11.3 $\pm$ 0.7
	Llama-70B	10.0 $\pm$ 0.3	10.8 $\pm$ 0.0	9.2 $\pm$ 0.3	1.0 $\pm$ 0.2	0.9 $\pm$ 0.4	0.7 $\pm$ 0.2	4.6 $\pm$ 0.2	5.0 $\pm$ 0.2	3.7 $\pm$ 0.2	5.2 $\pm$ 0.2	5.6 $\pm$ 0.2	4.6 $\pm$ 0.2
	Qwen-235B	10.6 $\pm$ 0.8	9.8 $\pm$ 0.3	9.2 $\pm$ 0.8	2.8 $\pm$ 0.6	2.8 $\pm$ 0.8	3.3 $\pm$ 0.8	6.4 $\pm$ 0.6	5.9 $\pm$ 0.6	4.4 $\pm$ 0.0	6.6 $\pm$ 0.6	6.2 $\pm$ 0.5	5.6 $\pm$ 0.5
TUA	Llama-8B	18.3 $\pm$ 3.3	17.7 $\pm$ 2.4	16.3 $\pm$ 3.4	9.9 $\pm$ 0.9	9.8 $\pm$ 1.5	9.3 $\pm$ 1.9	13.8 $\pm$ 1.3	13.3 $\pm$ 1.1	12.9 $\pm$ 1.3	14.0 $\pm$ 1.8	13.6 $\pm$ 1.7	12.8 $\pm$ 2.2
	Llama-70B	7.6 $\pm$ 1.2	8.0 $\pm$ 1.0	6.8 $\pm$ 1.6	5.2 $\pm$ 0.6	4.7 $\pm$ 0.9	4.7 $\pm$ 0.4	6.1 $\pm$ 0.2	5.5 $\pm$ 0.6	5.3 $\pm$ 0.4	6.3 $\pm$ 0.7	6.1 $\pm$ 0.8	5.6 $\pm$ 0.8
	Qwen-235B	10.2 $\pm$ 0.5	9.8 $\pm$ 0.8	8.4 $\pm$ 0.0	3.7 $\pm$ 0.6	4.1 $\pm$ 0.6	3.1 $\pm$ 0.0	7.1 $\pm$ 1.0	5.9 $\pm$ 1.2	7.3 $\pm$ 0.8	7.0 $\pm$ 0.7	6.6 $\pm$ 0.8	6.3 $\pm$ 0.3
MPA	Llama-8B	15.6 $\pm$ 0.1	14.4 $\pm$ 0.5	13.2 $\pm$ 0.3	11.6 $\pm$ 0.2	10.7 $\pm$ 0.2	10.2 $\pm$ 0.3	10.7 $\pm$ 0.2	9.8 $\pm$ 0.4	9.8 $\pm$ 0.4	12.6 $\pm$ 0.4	11.6 $\pm$ 0.3	11.1 $\pm$ 0.3
	Llama-70B	9.6 $\pm$ 0.3	10.2 $\pm$ 0.4	8.4 $\pm$ 0.1	2.7 $\pm$ 0.3	2.7 $\pm$ 0.1	1.3 $\pm$ 0.3	4.0 $\pm$ 0.2	4.0 $\pm$ 0.2	3.1 $\pm$ 0.1	5.4 $\pm$ 0.3	5.6 $\pm$ 0.2	4.3 $\pm$ 0.5
	Qwen-235B	12.7 $\pm$ 0.3	12.0 $\pm$ 0.2	9.6 $\pm$ 0.3	3.1 $\pm$ 0.1	2.7 $\pm$ 0.5	2.7 $\pm$ 0.3	7.6 $\pm$ 0.4	7.6 $\pm$ 0.1	7.1 $\pm$ 0.3	7.8 $\pm$ 0.3	7.4 $\pm$ 0.5	6.5 $\pm$ 0.3

Table 5.2: Flip rate (%) per model, backbone, perturbation type, and triage task (M=MANAGE, V=VISIT, R=RESOURCE). Values show mean $\pm$ std across 3 runs. Lowest overall per column are **bolded**. Shading indicates lowest flip rate per backbone group (same color-scheme as 5.1). “Average” reports the mean across perturbation types per task.

tent with prior work showing that LLM-generated summarization can alter treatment decisions even when clinical content is nominally preserved [35]. Gender Swap and Uncertain Tone consistently produce lower flip rates than summarization, suggesting models change decisions less frequently under word-level changes when compared to information compression.

Table 5.2 reports per-backbone results and provides evidence for **RQ1**. For Llama-8B, the baseline averages between 23–25% flip rate and augmented methods reduce this to 11.8–13.5%, a substantial reduction of about 50% for the best-performing method. Note that for Llama-8B, TUA and MPA reuse the tool plan produced by Llama-70B (Section 3.4.2), so their results reflect a mixed-backbone system (70B planner + 8B generator). MRA, which does not share this confound, achieves an approximately equivalent flip rate reduction (~48%), providing evidence that the gain is not purely attributable to the 70B planner.

For Llama-70B the gap narrows (7–9% baseline vs. 5–6.5% augmented, 35% rel-

ative reduction) but remains consistent across perturbation types. For Qwen-235B, the baseline is already near its lower bound at 6–7% and augmentation provides only marginal improvement (5.5–6.6% relative), most consistent across tasks on MRA. MPA on Qwen-235B averages slightly *above* baseline, indicating that tool-invocation variance exceeds the augmentation benefit at this capability level. This is also attributed to worse tool retrieval stability (Section 5.2.1) in Qwen-235B.

We apply Cochran’s Q test for matched binary outcomes to assess whether flip rates differ significantly across the four methods (see Section 3.5 for details); post-hoc pairwise comparisons use McNemar’s test with Bonferroni correction over six method pairs. The omnibus test is highly significant across all three perturbation types independently (Summary: $Q(3) = 88.3$, $p < 0.001$; Gender Swap: $Q(3) = 47.5$, $p < 0.001$; Uncertain Tone: $Q(3) = 90.7$, $p < 0.001$). Backbone-stratified tests confirm that the effect is driven by Llama-8B (all tasks: $Q(3) > 87$, $p < 0.001$) and is present but weaker for Llama-70B ($Q(3) = 15.6$, $p = 0.001$), while Qwen-235B shows no significant difference between methods ($Q(3) = 5.0$, $p = 0.17$). Augmentation therefore yields consistent gains for smaller backbones but provides no reliable benefit once the model is sufficiently capable.

In summary, knowledge augmentation improves robustness for smaller and mid-size backbones, but the effect diminishes as backbone capability grows. For Llama-8B, TUA and MPA gains reflect a mixed-backbone configuration (see Section 3.4.2). MRA, which retrieves paths without an LLM component, shows comparable gains supporting the conclusion that augmentation itself is beneficial. **Augmentation is most valuable when the model’s parametric clinical knowledge is weakest:** larger models already resist surface-level perturbations without external grounding. MPA achieves the lowest average flip rates for VISIT and RESOURCE on Llama-8B, while MRA is marginally lowest for MANAGE, consistent with KG paths providing more targeted support for management decisions where structured entity-relation reasoning is most relevant.

5.1.3 Reduced-Care Error Rate

RCER measures directional safety failure rather than overall consistency: it captures the proportion of perturbed cases where the model shifts to recommending *less* care than warranted, the clinically dangerous direction. Figure 5.2 reports RCER across 4 methods, 3 tasks, and 3 perturbations.

For **RQ1: for Llama-8B and Llama-70B, every knowledge-augmented method reduces RCER below the baseline on every task, indicating that augmentation improves robustness in the safety-relevant direction.** The largest absolute reductions are for

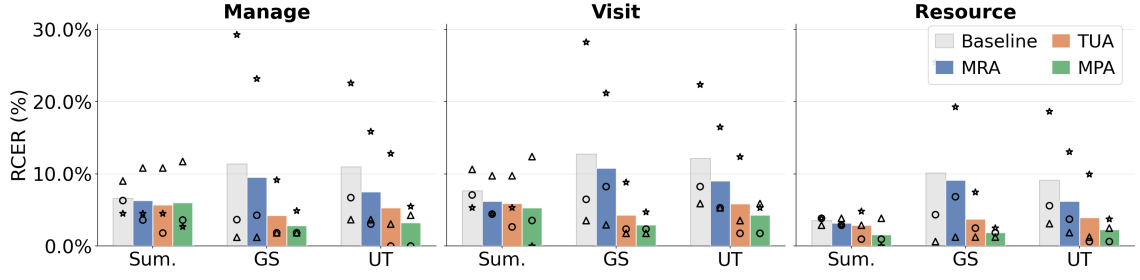


Figure 5.2: Reduced Care Error Rate (RCER) aggregated across the three backbone models. Higher RCER indicates more frequent clinically unsafe reductions in care. Here, • → Llama-3.3-70B, ▲ → Qwen3-235B-A22B, ★ → Llama-3.1-8B.

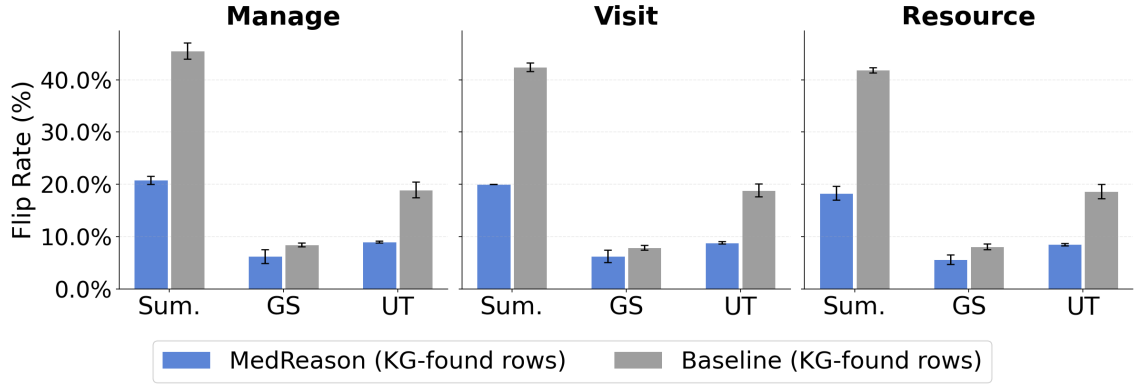


Figure 5.3: Impact of KG path availability on Llama-8B flip rate. Examples split by whether MRA retrieves at least one path, compared to the same rows from the Baseline.

Llama-8B, where the baseline averages 17.9% RCER; MPA reduces this to 3.3% (82% relative reduction), TUA to 6.2%, and MRA to 7.7%. For Llama-70B, the baseline RCER is lower (6.1%) but augmented methods reduce it to 1.0–1.4%. Qwen-235B shows no consistent reduction: baseline RCER is 4.6% and augmented method averages range from 4.0–5.1%, with MPA marginally *above* baseline (5.1%).

Perturbation-specific patterns reveal that Uncertain Tone and Gender Swap produce the highest RCER (11.1% and 11.3% averaged across methods and backbones respectively), while Summary, despite being the most disruptive for flip rate, produces lower RCER (6.1%). This dissociation occurs because summarisation perturbations shift models in both directions. By contrast, uncertain-tone and gender-swap perturbations more often shift predictions toward reduced care.

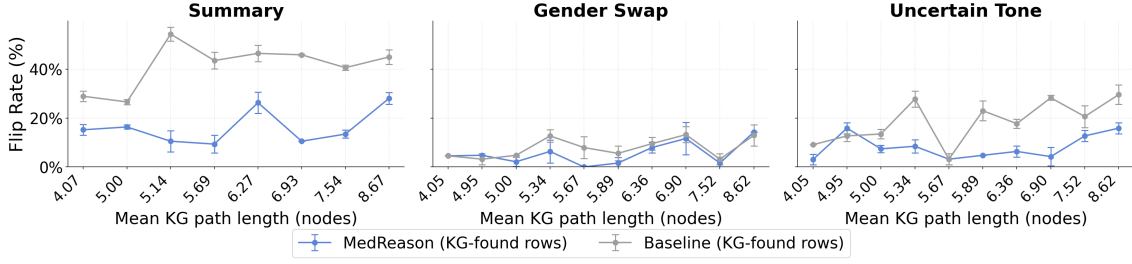


Figure 5.4: Flip rate (%) by mean KG path length for Llama-8B (MRA vs. Baseline). Standard deviation shown across 3 runs.

5.1.4 Effect of KG Path Coverage

We conduct further categorical analysis comparing MRA to the Baseline on Llama-8B, where augmentation gains are largest and where the practical case for KG grounding is strongest. This is because smaller models have weaker parametric clinical knowledge, making external structural guidance more consequential.

KG path availability. We partition MRA predictions by whether PrimeKG paths were found (the coverage is between 84–92% across perturbations). Figure 5.3 shows that MRA consistently lowers flip rate relative to the Baseline when paths are available, with reductions of approximately 30% for Summary, 10% for Uncertain Tone, and 3% for Gender Swap. Rows without retrieved KG paths show flip rates closer to Baseline levels, consistent with path availability driving the stability gain.

KG path length. Figure 5.4 plots flip rate against mean KG path length, binned into equal-frequency bins (8 bins for Summary, 10 for others). For Summary and Uncertain Tone, the Baseline shows persistently higher values as path length grows, whereas MRA remains comparatively closer to the first few. This suggests that KG-derived reasoning paths may help buffer against instability in clinically complex cases.

To quantify this directional pattern, we report upside semi-deviation (USD) adapted from the concept of semi-variance [60]. USD measures only increases in flip rate above the shortest-path reference bin Q_1 :

$$\text{USD} = \sqrt{\frac{1}{K-1} \sum_{i=2}^K \max(0, r_i - r_{Q_1})^2}, \quad \text{where } r_i \text{ is the flip rate in path-length bin } i, \\ \text{and } K \text{ is the total number of path-length bins.}$$

We use USD as a directional effect-size summary. A simple slope would test an approximately linear trend, while ANOVA would test for any difference between bins regardless of direction. By contrast, USD directly captures the quantity of interest here:

the extent to which instability increases relative to queries requiring short KG paths. Baseline USD is $2.5\times$ higher than MRA for Summary (16.26% vs. 6.44%) and $1.8\times$ higher for Uncertain Tone (13.23% vs. 7.36%). For Gender Swap, the difference is small (5.27% vs. 4.22%), consistent with this being a surface-level perturbation.

5.2 Are KG-Augmented Methods More Robust than Tool-Augmented Methods?

The results reveal a consistent dissociation between the two metrics that defines the RQ2 answer. On **flip rate**, MRA outperforms TUA under Gender Swap and Uncertain Tone across all three backbones, where the stability of KG retrieval translates into fewer decision changes. Under Summary, the ordering reverses: TUA achieves 11.5% versus MRA’s 13.4%. Section 5.2.1 examines a retrieval-based explanation for this reversal.

On **RCER**, the ordering inverts for the two backbone sizes where augmentation matters. For Llama-8B, TUA produces lower RCER than MRA; for Qwen-235B, TUA also beats MRA. Only for Llama-70B does MRA achieve lower RCER, and the difference is negligible. For the aggregate results across the models, TUA consistently reduces RCER further than MRA.

We hypothesize that this has a direct mechanistic interpretation. **KG paths stabilise which decision the model makes**; because the same entity-relation subgraph is presented for semantically equivalent inputs, the model is anchored to produce a consistent output regardless of surface variation. However, the paths do not always carry ground-truth clinical facts but they encode structural relationships between medical concepts. **Tool-retrieved factual evidence can correct the decision toward the clinically safer answer**: EuropePMC articles and FDA drug data provide specific clinical content that shifts the model away from under-allocation. Robustness (not changing) and safety (changing in the right direction) are therefore distinct properties, and KG and tool augmentation address them through distinct mechanisms.

5.2.1 Retrieval Stability Under Perturbation

The robustness differences observed in Table 5.2 can be partly explained by the stability of each method’s intermediate retrieval step, shown in Table 5.3. For each vignette, the same clinical scenario is presented in both its baseline and perturbed form; we compare what each method *retrieves* for the baseline version against what it retrieves for the perturbed version. Concretely, for MRA the retrieved content is the set of

Model	Backbone	Gender Swap			Uncertain Tone			Summary		
		JI	CosSim	Cov (%)	JI	CosSim	Cov (%)	JI	CosSim	Cov (%)
MRA	—	0.77	0.90	90.7	0.67	0.84	91.6	0.09	0.41	84.3
TUA	Llama-8B	0.39	0.71	99.6	0.25	0.58	99.6	0.17	0.49	100
	Llama-70B	0.45	0.72		0.28	0.60		0.15	0.44	
	Qwen-235B	0.36	0.61	87.6	0.26	0.52	91.1	0.13	0.38	88.6

Table 5.3: Retrieval consistency (JI and CosSim) between the content retrieved for the baseline vignette and the content retrieved for the perturbed vignette (see text for metric definitions). Coverage (Cov) gives the percentage of vignettes with non-empty retrieved content. MRA values are backbone-independent since KG path retrieval is deterministic and entity-anchored.

PrimeKG path strings returned by entity-anchored graph traversal; for TUA it is the document text summaries returned by the selected tools. We quantify agreement between these two retrieval outputs using two complementary metrics: word-level **Jaccard Index** (JI) [61], which measures exact token overlap between the baseline and perturbed retrieved texts, and biomedical-embedding **Cosine Similarity** (CosSim), computed via S-PubMedBERT-MS-MARCO [62] embeddings. JI captures whether the exact same tokens were returned while CosSim captures whether the content is semantically equivalent even when phrased differently, so together they distinguish surface retrieval instability (JI drops, CosSim holds) from deeper content divergence (both drop). Both metrics range from 0 to 1, where higher values indicate that perturbation did not change the evidence surfaced.

A key structural difference between the two methods is reflected in these scores: MRA retrieves KG paths by extracting explicit clinical entity mentions from the vignette and traversing PrimeKG, making retrieval anchored to clinical vocabulary. TUA must first plan which external tools to query before any evidence is retrieved, introducing an additional step where perturbations can alter behaviour.

Under both surface-level perturbations, MRA retrieves substantially more stable evidence than TUA. Path-set JI reaches 0.77 (Gender Swap) and 0.67 (Uncertain Tone), with CosSim of 0.90 and 0.84 respectively, reflecting that pronoun substitution and stylistic hedging leave the key clinical entity set intact, and therefore leave the retrieved KG subgraph largely unchanged. TUA achieves considerably lower consistency under the same perturbations, indicating that even when the underlying clinical facts are unchanged, small wording changes alter the tool-planning step and expose the downstream LLM to different retrieved content. This retrieval instability directly propagates into output flips: MRA with Llama-70B achieves the lowest Gender Swap flip rates across all models and backbones ($\leq 1.0\%$, Table 5.2), while MRA and MPA with

Llama-70B jointly achieve the lowest Uncertain Tone flip rates (4.0–4.6%), consistent with their high retrieval consistency under these perturbations.

The advantage reverses under summary perturbation. Summarisation compresses or removes entity mentions, causing MRA’s path Jaccard to collapse to 0.09 (exact-match rate: 3.6%), as missing or rephrased entities prevent graph traversal from reaching the same nodes. Despite this path-level collapse, CosSim remains at 0.41, indicating that the retrieved paths are broadly topically related even when specific edges differ. TUA degrades more gradually: content JI (0.13–0.17) and CosSim (0.38–0.49) are comparable to or exceed MRA’s, because keyword-based literature retrieval can still select relevant resources from a shorter query without requiring intact entity spans. This retrieval-level resilience is directly reflected in Table 5.2, where TUA Llama-70B achieves the lowest Summary flip rates across all models (7.6–8.0%), outperforming both KG-augmented methods on this perturbation type.

Retrieval **consistency** thus emerges as a key factor in both methods’ robustness profiles. When perturbations change the selected tools, they change the external evidence available to the LLM and increase the chance of inconsistent decisions. KG retrieval is more stable under surface-level demographic and stylistic perturbations because it is anchored to extracted clinical entities. It is, however, highly sensitive to perturbations that remove or compress those entities. Tool retrieval is less stable overall but degrades more gradually under summarisation, since keyword-based and dense retrieval can operate on shorter inputs without necessarily losing the primary query intent.

5.3 When Does MedPathAgent Outperform Other Methods?

Across the full perturbation suite, MPA achieves the highest joint Perfect Consistency on both Llama-8B (75.3%) and Llama-70B (88.6%, Table 5.1), indicating that the hybrid architecture provides stronger decision anchoring than either KG or tool augmentation alone. For per-perturbation flip rate (Table 5.2, Figure 5.1), results are more mixed: MPA achieves the lowest flip rates for Llama-8B under Summary and Llama-70B under Uncertain Tone, but does not consistently outperform MRA across all perturbation types. The tool-use component introduces variance that partially offsets the KG benefits, particularly under Gender Swap and Summary where tool retrieval is least stable.

On RCER, MPA achieves the lowest values consistently. The ordering $MPA < TUA < MRA < \text{Baseline}$ holds in 8 of 9 task-perturbation cells (exception: MANAGE under Sum-

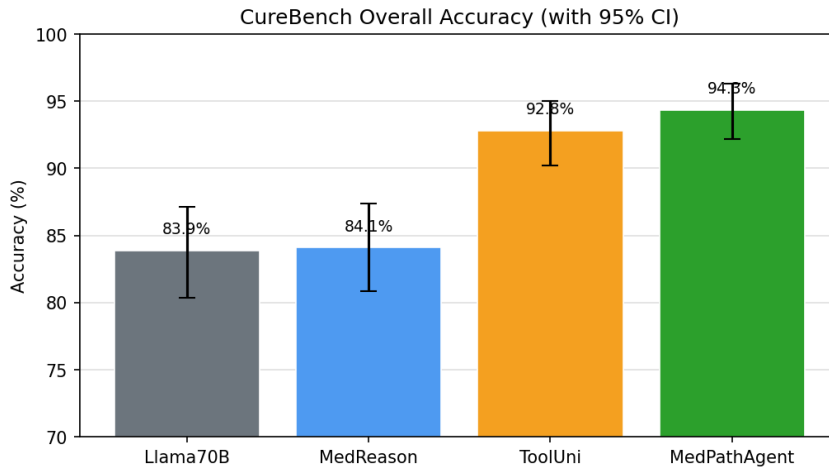


Figure 5.5: Accuracy on CureBench ($n = 459$) with bootstrapped 95% confidence intervals.

mary). The advantage is most pronounced under Uncertain Tone, where hedging language raises MANAGE RCER to 11.0% for Baseline but only 3.3% for MPA (recall that for MANAGE, a YES–self-manage response is the reduced-care direction, so RCER measures cases where perturbation incorrectly shifts the model toward recommending self-management), and on RESOURCE, where MPA achieves the widest margin (1.9% vs. 7.6% Baseline, 6.2% MRA, 3.5% TUA). Cases where MPA beats MRA correlate strongly with cases where TUA also beats MRA, consistent with the RCER advantage being driven primarily by the tool-retrieval component, with KG paths providing incremental targeting precision.

In summary, MPA reliably outperforms the Baseline on *safety* (RCER) across all conditions, and on *consistency* (PC, flip rate) primarily for the smaller backbones where augmentation is most effective. The following sections examine whether this advantage extends to factual accuracy on CureBench.

5.3.1 Accuracy on CureBench

The Baseline (Llama-70B) achieves 83.9%, already competitive for a zero-shot LLM. MRA achieves 84.1% (+0.2%), a negligible difference indicating that KG relational paths do not help factual MCQ tasks: PrimeKG encodes structural relationships, but CureBench requires precise label-level facts (exact dosages, interaction rates, contraindications) that are not graph-traversable. TUA achieves 92.8% (+8.9%, $p < 0.001$, McNemar’s test), consistent with the in-distribution advantage: the tools directly expose the answer content the dataset was generated from. MPA achieves 94.3% (+1.5% over TUA, $p < 0.05$) and this gain is consistent with KG-guided tool precision, where entity-relation

paths steer the planner toward a more specific FDA section than dense retrieval alone selects. The Baseline-to-TUA gap (8.9%) versus TUA-to-MPA gap (1.5%) reflects a retrieval ceiling: once the correct database is queried, additional structural guidance provides only incremental precision.

5.3.2 Qualitative Error Analysis

Case 1: brand-name entity resolution. *Q: Which of the following statements about G-62 is true regarding its solubility and absorption?* (A) high solubility, rapid (B) low solubility, slow (C) moderate (D) unpredictable. **Correct: B**

Baseline ✗	TUA ✓	MPA ✓
Chose D . “I’m not aware of any established medical literature on a substance called ‘G-62’. It’s possible that it’s a fictional or proprietary compound, or it may be a code name for a real substance. Without more information, I couldn’t find any reliable sources to provide an accurate answer.”	Chose B . “This is because G-62, also known as GaviLyte-G, is poorly absorbed , which suggests slow absorption. Although the text does not directly mention solubility, poor absorption often correlates with low solubility, making option B the most accurate description.”	Chose B . Identical reasoning to TUA (same OpenFDA brand lookup returned).

Case 2: KG-guided drug-interaction lookup. *Q: Why should GOPRELTO be avoided in patients taking disulfiram?* (A) reduces efficacy (B) increases plasma cocaine exposure (C) severe allergic reaction (D) decreased cholinesterase. **Correct: B**

Baseline ✗	TUA ✗	MPA ✓
Chose D . “ GOPRELTO (cocaine) is metabolized by plasma cholinesterase. Disulfiram inhibits this enzyme , which can lead to decreased metabolism of cocaine and increased risk of toxicity. Therefore, GOPRELTO should be avoided in patients taking disulfiram to prevent potential adverse effects.”	Chose D . “Based on general medical knowledge, disulfiram is known to inhibit certain enzymes, including aldehyde dehydrogenase and possibly others involved in drug metabolism . [...] <i>without specific information on ‘GOPRELTO’ [...] the best answer cannot be accurately determined without more context on ‘GOPRELTO.’</i> [...] option D might be considered, but with significant reservations.”	Chose B . KG path <i>disulfiram</i> → <i>cocaine (synergistic)</i> triggered a cocaine-specific query. “ disulfiram increases plasma cocaine exposure, including both AUC and C_{max}, by several fold after acute intranasal or intravenous cocaine administration [...] the increased plasma cocaine exposure caused by disulfiram makes it necessary to avoid GOPRELTO in patients taking disulfiram.”

Table 5.4: These examples show model reasoning for the methods which improved over baseline compared to Baseline qualitatively (Llama-3.3-70B) on the CureBench dataset. **Red** = reasoning that caused an incorrect response; **green** = retrieved knowledge that caused the correct response; **orange** = knowledge TUA correctly retrieved but found insufficient. These gaps were then addressed by MPA’s KG path.

Table 5.4 presents two worked examples with the original question text and answer options. Case 1 shows TUA correcting Baseline errors via direct FDA retrieval and Case 2 shows MPA additionally correcting TUA through KG-guided drug-interaction lookup. In both cases the error is a *knowledge access failure* rather than a reasoning failure.

Case 1 illustrates a knowledge access failure: the Baseline could not identify ‘G-62’

as a known drug entity, while TUA’s OpenFDA brand lookup resolved it to GaviLyte-G and retrieved the relevant pharmacokinetic label text. Case 2 illustrates a deeper failure where both Baseline and TUA applied a plausible enzyme-inhibition heuristic; TUA correctly identified that disulfiram inhibits enzymes but could not surface a cocaine-specific interaction without the PrimeKG edge *disulfiram* $\rightarrow$ *cocaine* (*synergistic interaction*), which directed MPA’s tool planner to the GOPRELTO drug-interactions section containing the explicit plasma cocaine exposure fact.

5.3.3 Effect of KG Path Coverage on CureBench Performance

Table 5.5 partitions accuracy by KG path availability. The 409 questions with paths involve entities well-represented in PrimeKG, while the 50 without paths likely involve post-2017 approvals or compounds where NER fails to extract a mappable entity. The Baseline gap between subsets (84.60% vs. 76.00%) reflects question difficulty rather than augmentation: established drug–disease questions are inherently easier, so this gap must be controlled for when comparing methods.

MRA achieves 84.84% on path-covered questions, only 0.24% above Baseline on the same subset. Even with paths available, the structural paths do not supply the label-level facts the MCQ requires, indicating that KG coverage is necessary but not sufficient for accuracy gains on lookup tasks. TUA achieves 93.15% with paths and 90.00% without, the 3% gap again reflecting difficulty rather than retrieval failure.

MPA achieves 94.87% on path-covered questions (+1.72% over TUA, 7 net recoveries out of 409), each corresponding to a case where a PrimeKG entity–relation path steered the planner to a more specific FDA section than dense retrieval alone selected. On the 50 uncovered questions, MPA falls back to TUA (90.00%) with no degradation, suggesting that KG conditioning usually improves performance and introduces no apparent negative transfer when paths are absent.

Table 5.5: Accuracy on the CureBench dataset split by KG path availability. MRA falls back to Baseline when no paths found; MPA falls back to TUA.

Model	Backbone	With KG Paths ($n = 409$)	Without KG Paths ($n = 50$)
Baseline	L70B-3.3	84.60%	76.00%
MRA	L70B-3.3	84.84%	<i>fallback to Baseline</i>
TUA	L70B-3.3	93.15%	90.00%
MPA	L70B-3.3	94.87%	<i>fallback to TUA</i>

Chapter 6

Discussion

6.1 Knowledge Augmentation and Robustness

Our results reveal a dissociation between two desirable properties of medical decision-support systems: *decision consistency* under input perturbation and *clinical safety* when predictions shift toward unsafe recommendations. MPA achieves the strongest results on both PC and RCER, particularly for smaller backbones, though per-perturbation flip rate results are more mixed across methods.

6.1.1 Retrieval Consistency as the Mechanism of Robustness

The central finding of this thesis is not merely that augmentation improves robustness, but *why* it does so, alongside why different augmentation types produce different robustness profiles. The retrieval stability analysis (Table 5.3) reveals that decision stability depends on whether the retrieval step returns equivalent evidence across perturbations. When MRA retrieves identical KG paths for a baseline and its perturbed variant, the downstream decision is anchored to the same relational context, and a flip becomes less likely. When paths diverge, as they do for 70% of Summary-perturbed cases (exact match rate 3.6%), this anchor is lost and flip rate rises toward Baseline levels.

The KG coverage analysis (Figure 5.3) is consistent with this within-method: on cases where no paths are retrieved, MRA’s flip rate is indistinguishable from the Baseline, consistent with path retrieval being the mechanism rather than backbone behaviour.

These findings suggest a preliminary design hypothesis, derived from this two-dataset three-perturbation evaluation: **in a clinically robust augmented system, the retrieval component should return consistent results across the input variation types expected in deployment.** KG retrieval satisfies this for surface-level stylistic and gender-substitution changes because NER extracts clinical entity mentions that are unaffected by pronoun substitution or hedged phrasing. Tool selection is less invariant to these changes but degrades more gradually under compression, since keyword-based and dense retrieval can operate on shorter inputs without losing the primary query intent. The choice of augmentation type should therefore be matched to the types of input variation expected in practice: settings where stylistic or gender-substitution variation is the primary perturbation type may benefit more from KG augmentation; settings where clinical summarisation is a primary input format may benefit more from tool augmentation.

6.1.2 Consistency vs. Safety

MRA consistently reduces flip rates across all backbones and perturbation types, with the largest gains under Gender Swap and Uncertain Tone. Structured KG paths act as invariant anchors: when surface-level linguistic features are altered, the biomedical relational structure from PrimeKG remains stable and guides the model toward a consistent decision. This consistency benefit does not, however, translate into a clinical safety advantage. MRA is the weakest augmented method on RCER, never achieving the lowest RCER on any task. A system can be highly consistent in the *wrong* direction. KG paths encode structural relational knowledge that stabilises decisions without providing the ground-truth factual content needed to correct them.

TUA achieves the largest accuracy gains on CureBench (+8.9%) and substantially reduces RCER relative to both the Baseline and MRA. The mechanism is factual grounding: EuropePMC articles and FDA drug data provide specific clinical evidence that shifts the model away from under-allocation. The aggregate RCER reduction relative to baseline is comparable between MRA and TUA, meaning the *safety* benefit of augmentation is not meaningfully differentiated by knowledge type, even though the *flip-rate consistency* benefit is (Cochran’s Q , $p < 0.001$ under Gender Swap; see Section 5.1). Robustness and clinical safety are therefore distinct properties requiring distinct evaluation, and a system that scores well on one cannot be assumed to score well on the

other.

TUA exhibits an apparent Visit-specific elevation in RCER: retrieved drug and adverse-event information appears to shift recommendations toward pharmacological monitoring over clinical contact, resulting in elevated RCER under Summary. The Llama-8B flip rate results also show that tool augmentation can *increase* instability under Gender Swap relative to baseline: tool-retrieved content may interact with gender-substituted surface features in ways that destabilise decisions for smaller models.

MPA achieves the highest accuracy on CureBench (94.3%) and the lowest RCER on all three tasks. The net question recoveries on KG-path questions are consistent with the hybrid mechanism: KG-guided entity retrieval improves tool precision on questions where the TUA planner fails to identify the correct FDA section. However, MPA does not uniformly outperform MRA on flip rate and inherits TUA’s VISIT vulnerability, reflecting a trade-off in hybrid architectures: retrieval increases factual grounding but also increases the number of intermediate steps over which perturbations can affect retrieval.

Our results provide a partial resolution: MPA’s RCER reductions on RESOURCE (1.9% vs. 7.6% Baseline) and MANAGE (3.3% vs. 11.0% Baseline under Uncertain Tone) demonstrate that KG-guided retrieval stabilises LLMs against perturbation-induced shifts toward under-allocation on the tasks where Gourabathina et al. [35] documents a conservative-care baseline, while MPA’s recovery of the VISIT RCER advantage over TUA shows that KG-conditioned tool selection can serve as a targeted corrective for a task-specific failure mode that uniform tool augmentation cannot address alone; the single exception (MANAGE under Summary) confirms that information-type-specific vulnerabilities are not fully eliminated even by hybrid augmentation.

6.2 Backbone Sensitivity

Backbone scale mediates the benefit of augmentation. For Llama-8B, augmentation produces the largest absolute *flip-rate* reductions (~50% relative). As noted in Section 3.4.2, TUA and MPA for Llama-8B use tool plans from Llama-70B; MRA, which does not share this configuration, shows comparable reductions (~48%), suggesting that smaller models benefit most from external structural grounding when their parametric knowledge is insufficient to maintain stable decisions under paraphrase, and that this benefit is not solely attributable to the larger planner. Cochran’s Q backbone-stratified tests confirm this for *flip rate*: the effect is highly significant for Llama-8B ($Q(3) > 87$, $p < 0.001$ across all tasks; see Section 5.1) but not for Qwen-235B ($Q(3) = 5.0$, $p = 0.17$). For *RCER* the pattern inverts: the safety benefit is concentrated at Llama-70B over

Llama-8B, likely because the smaller number of care-warranted cases in the Llama-8B subset reduces statistical power for RCER comparisons at this backbone size. KG augmentation is therefore most valuable for consistency in settings where computational constraints require smaller models, while the safety benefit is most pronounced for mid-size backbone models.

This result is consistent with Gao et al. [12], whose 8B fine-tuned model outperforms the 70B Llama and GPT-4o by 39.3% and 25.8% respectively on open-ended drug reasoning by substituting parametric recall with verified tool retrieval, demonstrating that a sufficiently capable tool-use layer allows smaller models to match or exceed larger ones on tasks where parametric knowledge is the bottleneck.

6.3 Task Heterogeneity

The three triage tasks are not equally sensitive to augmentation strategy. **RESOURCE** benefits most consistently from combined KG and tool augmentation; **MANAGE** also benefits across most conditions, with the exception of the Summary perturbation where the MPA ordering does not hold. Both decisions hinge on drug-specific factual information that PrimeKG and ToolUniverse both encode. **VISIT** decisions require integrating patient presentation with broader clinical escalation judgement, a task for which retrieved pharmacological content systematically miscalibrates. **VISIT** is the task where MPA most reliably prevents reduced-care errors relative to the Baseline, yet also the task where TUA’s retrieval content most frequently misdirects. This apparent contradiction reflects a compositional effect: MPA’s KG-guided tool selection recovers the Visit safety benefit that TUA alone cannot sustain.

This task-level sensitivity is consistent with both MedPerturb [35], which reports that the three triage tasks (Manage, Visit, Resource) show distinct sensitivity patterns across perturbation types. LLMs systematically over-allocating resources and under-recommending self-management in ways that differ by task. Yan et al. [38] showed that different types of perturbed clinical information (gender, age, symptoms, checkup results) affect diagnostic decisions differently, supporting the conclusion that task-aware retrieval is needed over uniform augmentation. The implication for architecture design is discussed in the Future Work section below.

6.4 Future Work

Internal Reasoning Methods We do not compare the performance of our methods to advanced internal reasoning prompting methods such as Chain-of-Thought [63],

Chain-of-Verification [64] or Self-Consistency [65]. We also do not consider multi-agent internal approaches such as MedAgents [39]. These were beyond our scope since our key focus was on comparing knowledge-augmented approaches but these could be included in future work for more comprehensive analysis of external knowledge benefits. We refer the reader to knowledge augmentation literature (Chapter 2) to understand the preference for knowledge-augmentation over internal reasoning in the biomedical reasoning domain.

Extending the retrieval invariance analysis. A key finding is that intermediate retrieval consistency predicts decision consistency. This opens a direct research direction: designing retrieval components that are explicitly trained to be perturbation-invariant. For KG retrieval, this could involve perturbation-augmented NER fine-tuning so that entity extraction is robust to style and compression changes. For tool retrieval, it suggests training ToolRAG on perturbation-paired queries so that tool selection is stable across surface variants of the same clinical question.

Broader evaluation benchmarks. The current evaluation covers MedPerturb robustness and accuracy on CureBench. Future work should evaluate on MedXpertQA [37] and MedR-Bench [47], which assess expert-level reasoning quality and multi-stage clinical case reasoning respectively, to determine whether the robustness-accuracy trade-offs observed here generalise to harder tasks. Yan et al. [38] provides a complementary robustness benchmark, DiagnosisQA which focuses on clinically meaningful perturbations. This would help us test whether the retrieval invariance effect holds for perturbations with clinical consequence.

Task-aware retrieval. The VISIT miscalibration observed for TUA suggests that pharmacological retrieval content is not uniformly beneficial across triage decisions. A routing layer that classifies the decision type before retrieval could help. Suppressing drug-specific tools for clinical contact decisions, for instance, could recover TUA’s RCER disadvantage on VISIT while preserving its accuracy gains on MANAGE and RESOURCE.

Smaller knowledge graphs. PrimeKG (~4M edges) requires significant memory and preprocessing. Evaluating lighter alternatives, such as disease ontologies, SNOMED CT subsets, or dynamically constructed per-query subgraphs, would determine whether the robustness benefit is contingent on PrimeKG’s coverage or whether smaller structured resources can achieve comparable retrieval invariance.

Chapter 7

Conclusion

This thesis evaluates knowledge-augmented medical reasoning agents on decision robustness under non-clinical perturbation, clinical safety, and pharmaceutical question answering accuracy.

RQ1: Are knowledge-augmented methods more robust to non-clinical perturbations than pure LLM baselines?

Yes, for Llama-8B and Llama-70B. Every augmented method reduces both flip rate and RCER relative to the pure LLM baseline; Llama-8B baseline flip rates of ~23–25% fall to 11–14% with augmentation. MRA achieves comparable reductions without the mixed-backbone confound present in TUA and MPA for Llama-8B (see Section 3.4.2), supporting the general conclusion. For Qwen-235B the baseline is already highly stable and augmentation provides no reliable benefit.

RQ2: Are KG-augmented methods more robust than tool-augmented methods?

The answer depends on the metric. KG augmentation (MRA) produces stronger *flip-rate* consistency, particularly under gender-swap and uncertain-tone perturbations where PrimeKG paths anchor decisions to stable relational structure. Tool augmentation (TUA) produces lower *RCER*: factual retrieval corrects decisions toward the safer answer in a way that structural paths alone do not. KG augmentation improves consistency; tool augmentation improves safety—distinct properties requiring separate evaluation.

RQ3: Does combining KG-guided reasoning with tool-augmented retrieval outperform either strategy in isolation, and under what conditions?

MPA achieves the lowest RCER on all three triage tasks and the highest accuracy on CureBench (94.3%, +1.5% over TUA, $p < 0.05$), with KG paths conditioning tool planning toward more specific pharmaceutical entities. However, the hybrid does not uniformly dominate MRA on flip rate and inherits TUA’s VISIT vulnerability: the additional retrieval steps increase the surface area over which perturbations can propagate.

Limitations. CureBench results are partly inflated for ToolUniverse-based methods, as the tools directly expose content from the same database used to generate the benchmark; out-of-distribution evaluation on MedXpertQA [37] and MedR-Bench [47] is needed to test whether gains generalise. The VISIT vulnerability warrants task-aware retrieval gating. The demographic evaluation covers a single axis (binary gender substitution); whether findings extend to other demographic axes is unknown.

Contributions. We demonstrate that knowledge augmentation improves decision robustness under non-clinical perturbation for smaller and mid-size backbones, with backbone capability the primary moderator of benefit. We show that KG and tool augmentation address distinct properties—decision consistency and clinical safety respectively—and that comprehensive robustness evaluation requires measuring both flip rate and clinically directional error separately. We introduce TUA, a compute-tractable TxAgent-style implementation that enables multi-backbone evaluation across the four methods studied here. We introduce MPA, a hybrid agent that uses PrimeKG reasoning paths to condition pharmaceutical tool planning, achieving the lowest RCER and highest accuracy on CureBench among evaluated systems on the Llama-70B backbone.

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